

Education

2025 – 2027 Expected	Carnegie Mellon University , Pennsylvania, USA M.S. in Robotics · GPA 4.17/4 · Advisor: Changliu Liu
2021 – 2025	Shanghai Jiao Tong University , Shanghai, PRC B.Eng. in Automation · GPA 3.92/4

Motivation

As learned end-to-end models become increasingly capable in planning and control, my research asks what structure robots should still represent explicitly. I study how to ground such structure in experience and compile it into task-aware representations, such as memory and constraints, to inform interpretable and compositional decision-making.

Selected Publications

* Equal contribution

RSS 2026	[1]	Emergent Neural Automaton Policies: Learning Symbolic Structure from Visuomotor Trajectories Yiyuan Pan*, Xusheng Luo*, Hanjiang Hu, Peiqi Yu, Changliu Liu
NeurIPS 2025	[2]	Seeing through Uncertainty: Robust Task-Oriented Optimization in Visual Navigation Yiyuan Pan, Yunzhe Xu, Zhe Liu, Hesheng Wang
NeurIPS 2025	[3]	Wonder Wins Ways: Curiosity-Driven Exploration through Multi-Agent Contextual Calibration Yiyuan Pan, Zhe Liu, Hesheng Wang
AAAI 2025 Oral	[4]	Planning from Imagination: Episodic Simulation and Episodic Memory for Vision-and-Language Navigation Yiyuan Pan, Yunzhe Xu, Zhe Liu, Hesheng Wang
CoRL 2026	[5]	Inference-Time Robot Behavior Steering through Physically-Aware Reconfiguration of Task-Structure Yiyuan Pan, Hanjiang Hu, Shangtao Li, Xusheng Luo, Changliu Liu
AAAI 2025 Oral	[6]	FLAME: Learning to Navigate with Multimodal LLM in Urban Environments Yunzhe Xu, Yiyuan Pan, Zhe Liu, Hesheng Wang
TPAMI 2026	[7]	Dream to Recall: Imagination-Guided Experience Retrieval for Memory-Persistent Vision-and-Language Navigation Yunzhe Xu, Yiyuan Pan, Zhe Liu
IROS 2026	[8]	Learning Safe-Stoppability Monitors for Humanoid Robots Yifan Sun, Yiyuan Pan, Shangtao Li, Caiwu Ding, Tao Cui, Lingyun Wang, Changliu Liu
ICLR 2026	[9]	Seeing, Listening, Remembering, and Reasoning: A Multimodal Agent with Long-Term Memory Lin Long*, Yichen He*, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, Wei Li
arXiv 2026	[10]	VLESA: Vision-Language Embodied Safety Agent for Human Activity Monitoring Hanjiang Hu*, Yiyuan Pan*, Jiaying Li, Xusheng Luo, Alexander Robey, Na Li, Yebin Wang, Changliu Liu

Experience

(i) Contextualizing Decisions: What to Remember

Sep 2023 –	Shanghai Jiao Tong University <i>Advisor: Hesheng Wang</i>
May 2025	• Adapted a multimodal language model for urban navigation using cross-attention over recent observations and three-stage tuning on scene descriptions, route summaries, and actions. [6]

- Fused observations with recursively imagined scenes in topological memory [4]; queried environmental and behavioral memories across episodes with language-conditioned world-model predictions [7].
- Encoded interaction histories as graphs of inferred peer intentions to calibrate intrinsic rewards, separating informative novelty from environmental stochasticity in multi-agent exploration. [3]

Mar 2025 –

ByteDance Seed *Advisor: Hang Li*

Jul 2025

- Designed entity-centric multimodal memory that incrementally links episodic events and semantic knowledge via persistent identities, supporting cross-modal retrieval and multi-turn reasoning. [9]

(ii) Assessing Decisions: What to Follow

Jun 2024 –

California Institute of Technology *Advisor: Steven Low*

Dec 2024

- Developed a learning-to-optimize framework that maps visual predictions into calibrated convex uncertainty sets for robust navigation, using planning feedback to align network training with downstream decision quality rather than prediction accuracy alone. [2][†]

Sep 2025 –

Carnegie Mellon University *Advisor: Changliu Liu*

Present

- Learned a neural monitor estimating safe-stopping probability under a fixed fallback policy, using importance sampling near the safety boundary to reduce false-safe predictions on humanoids. [8]
- Coupled intent-action prediction from egocentric video with a goal-conditioned safety filter trained through reinforcement learning, evaluating anticipated human actions against context-dependent safety requirements to support proactive intervention. [10]

(iii) Organizing Decisions: What to Symbolize

Sep 2025 –

Carnegie Mellon University *Advisor: Changliu Liu*

Present

- Learned state-machine task models from visuomotor demonstrations through adaptive clustering and automata inference, without symbolic annotations; coupled their action priors with a residual policy for phase-conditioned action generation. [1]
- Composed user-preference automata with learned task structure and recomputed action priors through demonstration replay, grounding symbolic edits in demonstrated behaviors while steering a frozen residual policy at inference time. [5]

Mar 2026 –

University of Pennsylvania *Advisor: Jiayuan Mao*

Present

- Formalized solver-sufficient symbols that expose discrete commitments and leave efficiently solvable subproblems; developed counterexample-guided refinement to discover compact symbol sets while preserving optimality in structured hybrid models. *[under review]*
- Extended symbol discovery to neural energy landscapes using sampled optimization trajectories to identify barriers and refine partitions, enabling region-wise optimization and behavior steering without retraining. *[under review]*

Technical Skills

Programming	Python, C/C++, MATLAB
Machine Learning	PyTorch, Transformers, OpenAI / Azure API, W&B
Robotics	Kinova Gen3, Unitree G1, Galaxea R1 Lite · MuJoCo, Isaac Lab, Habitat-Sim
Formal Methods	Convex / combinatorial optimization, automata theory, temporal logic

[†] The framework began as my Caltech SURF report on power systems and was later instantiated as a robotics paper at SJTU.